A giant crayfish contest leads scientists to a growing mystery

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It started with a little friendly bragging. **Biologist** Caitlin Bloomer pulled a huge crayfish from the mud during a survey in Kentucky. Its species is actually called the Crawzilla crawdad, and she was sure it was the biggest crayfish anyone had ever seen.

When she showed it to a colleague, he took one look and said, "No way, I've caught bigger," Bloomer recalled. Bloomer teaches at the University of Illinois Urbana-Champaign.

That challenge turned into a nationwide search. Bloomer asked **biologists** and museum **curators** across the country to check their collections for unusually large crayfish. The team used **calipers**, tools that measure objects very precisely, to compare preserved **specimens**. They found 20 extra-large crayfish from 18 different species. The results were published in the scientific journal Ecology.

Bloomer's Crawzilla made the top 20. "I'm super proud of it," she said.

The winner was a Tennessee bottlebrush crayfish kept at West Liberty University in West Virginia. It was collected in 2011, and its body measured 86.6 millimeters, about 3.4 inches, not counting its tail, claws and antennae. Professor Zackary Graham said its hard upper shell alone is about the length of a credit card.

The search also raised a scientific puzzle. The largest crayfish of many species all seemed to stop growing at about 80 millimeters. Yet some crayfish elsewhere, such as the Tasmanian giant freshwater crayfish, grow much larger.

The researchers checked whether location, oxygen levels or **habitat** type could explain the size limit, but they found no clear pattern. Instead, the limit may come from the animals' own bodies. Their hard outer shells, called **exoskeletons**, or their muscles might set a physical limit on how big North American crayfish can grow.

Word Treasure

biologist -- A biologist is a scientist who studies living things.

curators -- Curators are people who take care of and study the objects in a museum's collection.

calipers -- Calipers are tools that measure the size of an object very precisely.

specimens -- Specimens are samples of plants or animals kept so scientists can study them.

exoskeletons -- Exoskeletons are hard outer coverings that protect the bodies of animals like crayfish and insects.

habitat -- A habitat is the natural place where an animal or plant lives.

Background you need

Crayfish, also called crawdads, are freshwater animals related to lobsters and shrimp.

Like other animals with exoskeletons, a crayfish must shed its hard outer shell and grow a new one in order to get bigger.

Museums and universities keep preserved animal specimens so scientists can compare and measure them many years later.

Scientists studying maximum size look at things like where an animal lives and how its body is built to find out what limits its growth.

Break it down

LEAD: A friendly brag about a giant crayfish led to a scientific discovery.
KEY EVENT: A nationwide search of museum and university collections
+- WHO: Caitlin Bloomer of the University of Illinois Urbana-Champaign and other biologists and curators
+- HOW: Measuring preserved crayfish with calipers; results in the journal Ecology
RESULTS: 20 extra-large crayfish from 18 species; the record was a Tennessee bottlebrush crayfish at 86.6 millimeters
PUZZLE: Many species seem to max out at about 80 millimeters
EVIDENCE: Location, oxygen and habitat showed no clear pattern
OPEN QUESTION: Do shells or muscles set the limit?

Why it matters

Learning what limits an animal's size helps scientists understand how bodies are built and how old museum collections can still answer new questions.

Test what you remember

1. Where did Caitlin Bloomer find her giant crayfish?

- (A) In a museum in Illinois
- (B) In a river in Tasmania
- (C) In the mud during a survey in Kentucky
- (D) In a lake in West Virginia

2. What started the nationwide search for big crayfish?

- (A) A museum lost its collection
- (B) A colleague said he had caught bigger
- (C) A new law about crayfish
- (D) A government contest

3. What are calipers?

- (A) A type of crayfish
- (B) Tools that measure objects very precisely
- (C) Nets for catching animals
- (D) Museum display cases

4. Which crayfish was the biggest?

- (A) A Tennessee bottlebrush crayfish at West Liberty University
- (B) A burrowing crayfish from Illinois
- (C) Bloomer's Crawzilla
- (D) A Tasmanian giant freshwater crayfish

5. At about what size did the largest crayfish of many species seem to stop growing?

- (A) 20 inches
- (B) 80 millimeters
- (C) 3.4 meters
- (D) 8 millimeters

6. What might set the size limit, according to the researchers?

- (A) Water temperature only
- (B) Fishing by people
- (C) The animals' shells or muscles
- (D) The amount of sunlight

Answer key (try the quiz first!): 1: C 2: B 3: B 4: A 5: B 6: C

Questions to think about

MY THOUGHTS (at least 20 words):